

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

AU-5002-CBGS

B.E. V Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Measurement And Control

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1.
 - a) Describe with examples the applications of measurement systems.
 - b) Explain about the following static characteristics:
 - i) Accuracy
 - ii) Sensitivity
 - iii) Range
 - iv) Precision
2.
 - a) Draw diagrams to show how LVDTs (Linear Variable Differential Transducers) can be used with bellows elements and Bourdon tubes for measurement of pressure. Give their advantages and disadvantages.
 - b) What is the working principle of Orifice meter? Discuss with a diagram.
3.
 - a) What is calibration? Discuss its importance.
 - b) What do you mean by sensitivity and range of a measuring instrument?
4. State the different types of errors and explain each in brief.

AU-5002-CBGS

PTO

[2]

5. A study has indicated that the life of TV picture tubes manufactured by a certain firm is normally distributed with a mean life of 5 years (1 year = 365 days) and a standard deviation of 500 days. The manufacture gives a guarantee of 1 year. Determine
- What percentage of picture tubes will he have to replace in 1 year?
 - If the manufacturer wishes to replace the same amount of picture tubes with 2 years guarantee, what should he do?
6. a) What is uncertainty analysis? How it is done?
b) Write about data acquisition and data reduction.
7. a) Explain open loop and close loop control system.
b) Discuss first and second order systems.
8. Write short notes on the followings:
- Principle of working of thermocouple
 - Transfer function
 - Rota-meter
 - Piezo-electric load cells
